Worksheet Week 9 #1 – Objects
EC327 Introduction to Software Engineering – Spring 2026

Label the following in the code snippet: Default constructor, constructor, public member variables, private member variables, and public member functions.

```
class Whale
{

     public:

     double length;

     Whale()
     {
          blubber = 100;
          length = 110;
          species = “Blue”;
     }

     Whale (int b)
     {
          blubber = b;
     }


     void eat(int food)
     {
          blubber += food/10;
     }

     int getBlubber()
     {
          return blubber;
     }

     private:
          int blubber;
          string species;
};
```

Draw a UML class diagram for the Whale class code snippet

Provide a **non-member function** to overload the “+” operator for two whale objects where whale1 + whale2 => returns a whale with blubber = whale1.blubber + whale2.blubber.

Provide a **member function** to overload the “/” operator for whale which returns a whale of length = whale1.length/ whale2.length.

Why make overloaded operators members vs. non-members?

What is a const member function? Write one for the Whale object.

What is the binary scope resolution operator? Write the “eat” function using this binary scope resolution operator so that you can define the function outside of the class.

What is the “this pointer”? What is an example of function which could use this in the Whale class?

What is a static member variable? Add a static member variable *int numberofwhales.* What do you need to add to the constructor update this properly?

What is a “destructor”? How many destructors can a class have? Write a destructor for a Whale.

What is a copy constructor? What is a deep copy vs. shallow copy?